

Ranking Transfers, Thresholds Don’t: Prior-Corrected Test-Time Adaptation for Audio Deepfake Detection

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Abstract—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show the failure is one of *operating point*: a fine-tuned self-supervised detector’s *ranking* (ROC-AUC) largely transfers cross-corpus while its decision *threshold* does not, so unlabeled target audio can repair the operating point at test time. We then show that confident-quantile test-time adaptation (TTA) can be sound only while its tail budget satisfies $q \leq \min(\pi, 1-\pi)$: a counting bound caps pseudo-label purity at $\min(1, \pi_{\text{real}}/q)$ for any detector. That holds where TTA is measured and essentially nowhere else — ASVspoof2021-DF is 97.2% spoof, In-the-Wild 37.2% — so at the published $q=0.3$ no model can exceed 9.3% purity on DF. Empirically, on the official DF eval (400,435 trials) that configuration destroys the operating point of three released state-of-the-art checkpoints, driving accuracy from 98.8–99.1% to 62–65%; on the one where we also ran the published TTA baselines, three of four degrade it as well, each through its own version of the assumption. Correcting the budget with a label-free prior estimate ($\hat{\pi}=0.9715$ vs. a true 0.9708) restores all three and beats the symmetric budget on every one; both EER and AUC move, so no re-thresholding rule accounts for it. The correction’s boundary we measure rather than assume: the prior is not identifiable from a model’s own outputs, so its estimate degrades exactly as the operating point does ($r=0.998$), and a wrong prior is worse than none. Since q is ours to choose and π is not, shrinking q satisfies the condition without estimating anything.

Index Terms—audio deepfake detection, anti-spoofing, test-time adaptation, label shift, cross-corpus generalisation, EER.

I. INTRODUCTION

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [17].

We start from a diagnostic measurement: a fine-tuned XLS-R detector evaluated on an unseen corpus keeps a high ROC-AUC while its accuracy at a fixed threshold collapses. It still *rank*s clips correctly; its operating point is mis-calibrated. Because ranking transfers, the most-confident predictions on unlabeled target audio are trustworthy enough to adapt on at test time.

That observation is the starting point of a family of methods, not a method. The step every member of the family takes—

label the extreme quantiles of the target scores and self-train on them—carries an assumption that is invisible on the benchmarks the field uses and fatal on the pools it deploys to: that the target pool is roughly class-balanced. Take the top and bottom 30% of a 97%-spoof pool and the “confidently bona fide” bucket is mechanically made of low-scoring *spoof*. Self-training then teaches the detector that a great deal of spoof is genuine. This is not hypothetical: it is what happens to all three released state-of-the-art checkpoints we test on the official ASVspoof2021-DF protocol (Table I), and to three of four modern TTA baselines.

Contributions. (i) The ranking–calibration asymmetry as the mechanism of cross-corpus failure, over 369 third-party cells and tested directly: a proxy $\mathcal{A}$ -distance *rises* after adaptation, so the effect is operating-point repair, not learned invariance. (ii) The class-prior assumption hidden in confident-quantile TTA and in three of four modern TTA baselines, measured on the field’s standard benchmark at its real prevalence, where it destroys the operating point of three released SOTA checkpoints. (iii) A label-free correction—shift estimation of the target prior, used to split the pseudo-label budget—which removes the damage on every DF checkpoint, threshold-free, together with the identifiability argument that bounds when it can be trusted. (iv) Two label-free monitors that separate damaging adaptations from safe ones, including our own published configuration.

Related work. SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]. Train-time domain generalisation aligns source domains [5] or runs gradient reversal over source and target jointly [13]; both act before deployment. In TTA, entropy minimisation (Tent) [7] is the reference recipe, and its instability prompted reliable-sample filtering (ETA/EATA) [8], sharpness-aware entropy with a recovery scheme (SAR) [9], and information maximisation with self-labelling (SHOT) [10] — we compare against all four. Batch-statistics recalibration [14] is a no-op for LayerNorm. The prior estimator is black-box shift estimation [11]; the proxy $\mathcal{A}$ -distance is from [15].

II. METHOD

Let x be a waveform with label $y \in \{0, 1\}$ (fake= 1) and let f_θ produce $s_\theta(x) = P_\theta(y=1 \mid x)$. We are given a source

model, its labelled training corpus $\mathcal{S}$, and an *unlabelled* target pool $\mathcal{T}$. No target label is read at any point.

What breaks under corpus shift. Ranking quality $\text{AUC} = \Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free; accuracy at a fixed τ is not. Empirically AUC degrades far less than accuracy at $\tau=0.5$ under source→target shift: across 369 third-party evaluation cells the accuracy a model’s ranking supports but its shipped threshold does not deliver averages 7.29 points. The deployable object is therefore the score *order*, and the broken object is the operating point. Everything below follows from taking that literally.

(a) The target prior, label-free. An operating point is a statement about a prior, so we estimate the prior rather than assume it. Black-box shift estimation [11] gives $\hat{p} = M^{-\top} q$, where $M[y, \hat{y}] = P_{\mathcal{S}}(\hat{y} | y)$ is the source confusion matrix measured on the model’s own labelled training data and q is the observed target prediction histogram. Estimators split into two families and the split matters: BBSE and EM [12] read the model’s *thresholded prediction rate*, a two-component Gaussian mixture reads the *shape* of the score histogram. The first inherits whatever calibration error the target induces — on In-the-Wild, where the checkpoint predicts 58.9% fake on a 37.2%-fake pool, BBSE returns 0.600 and EM 0.614 while the mixture returns 0.422; on the well-calibrated DF pool all three agree to within 0.02 (Sec. V). We report DF with BBSE as run, a mixture prior giving a near-identical budget there.

(b) The tail budget, and when it is valid. The standard rule takes the bottom q and top q of the target scores as pseudo-real and pseudo-fake. Its validity has a characterisation that needs no model at all. A bucket of $q_c N$ clips can contain at most the pool’s $\pi_c N$ class- c clips, so *for any detector whatsoever*

$$\text{purity}_c \leq \min(1, \pi_c / q_c). \quad (1)$$

This is counting, not an approximation; what is empirical is its *tightness*, which is where the ranking precondition enters — across all seven (checkpoint, corpus) cells here the bound is never violated and is attained to within 0.025, the slack tracking AUC (0.000 at AUC .985, 0.025 at .917). Two consequences follow, and they are the whole story.

The *symmetric* rule sets $q_c=q$ and is therefore sound only while

$$q \leq \min(\pi, 1 - \pi). \quad (2)$$

In-the-Wild satisfies this at $q=0.3$ ($\min=.372$) and the recipe repairs checkpoints there; DF does not ($\min=.028$), where no detector can exceed 9.3% pseudo-real purity — we measure 8.6% — and the recipe destroys them. The damage is one-sided: DF’s pseudo-fake tail is 100% pure at every q .

The *prior-split* rule, $q_c=2q\hat{\pi}_c$, restores Eq. (1) for *any* $q \leq \frac{1}{2}$ given a correct prior, and is identical to the symmetric rule at $\hat{\pi}=0.5$. With an estimate it needs

$$\hat{\pi}_c / \pi_c \leq 1/(2q) \quad \text{for both classes}, \quad (3)$$

i.e. at $q=0.3$ the estimate may overstate either class by at most $1.67\times$. This is why it works on DF ($\hat{\pi}_{\text{real}}/\pi_{\text{real}}=1.03$) and fails on In-the-Wild s42, where underestimating π_{real} inflates

the *fake* budget to 0.587 against $\pi_{\text{fake}}=0.372$ ($2.63\times$, bound 0.633) and simply moves the impurity to the other tail. Across the seven adapted arms we ran, the worst-tail bound is 1.000 for the five that came out intact (82–99% accuracy) and 0.093 and 0.633 for the two that collapsed (62.96, 47.99). Since q is ours to choose and π is not (Sec. V), a third option is to shrink q until Eq. (2) holds for the most skewed prior anticipated — at a supervision cost the prior-split budget avoids ($q=0.02$ labels 800 of 20,000 clips against 12,000). A fixed confidence threshold is a fourth, prior-free but calibration-dependent. We report all four.

(c) Objective. With pseudo-labels $\tilde{y}$ from (b) on the confident set $\mathcal{C}$ and a stochastic channel perturbation $a(\cdot)$ (per-clip gain $g \sim \mathcal{U}(0.7, 1.3)$ plus Gaussian noise $\sigma=0.005$),

$$\min_{\{\gamma, \beta\}, \phi} \underbrace{\sum_{x \in \mathcal{C}} \text{CE}(f_\theta(x), \tilde{y})}_{\mathcal{L}_{\text{ST}}} + \lambda \underbrace{\|f_\theta(a(x)) - \text{sg}[f_\theta(x)]\|^2}_{\mathcal{L}_{\text{cons}}}, \quad (4)$$

optimising only the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned blocks and the head ϕ (16,706 parameters), with $\lambda=0.3$, $q=0.3$. Restricting the trainable set and using only prior-corrected confident tails is what prevents the degenerate collapse to which entropy minimisation is prone.

(d) Two label-free guards. Adaptation can damage the model it is given, so a deployer needs to decide — without labels — whether to keep the adapted weights. Since the source ranking is the asset being exploited, we compare the adapted scores’ order against the frozen source model’s, $\rho = \text{Spearman}(s_\theta(\mathcal{T}), s_{\theta_0}(\mathcal{T}))$, and separately track the drift of the predicted positive rate from $\hat{\pi}$. We use these *post hoc*, as a keep/abort decision on a completed adaptation; a per-step stopping rule is the natural extension and we do not claim it here.

III. EXPERIMENTAL SETUP

Protocol A (published baselines exist). The official ASVspoof2021-DF eval partition against the official CM keys; 400,435 of 611,829 trials are held locally, the missing part uniform across codec, label and phase (verified 75–25 in every stratum), true spoof rate 97.22%. For scale, published pooled EERs on this protocol run from 1.95 (Wav2DF-TSL) and 2.87 (SSL+SLS) through 8.54 (Wav2Vec2-AASIST) to 15.64 (challenge DF top-1); our own source model reaches only 26.33, so we run the method on *released* checkpoints instead: DeepFense XLS-R 300M + AASIST trained on ASVspoof2019-LA [22], at three independent training seeds. Adaptation uses an unlabelled 20,000-clip subsample of the eval pool; we report the full official pooled EER and, as an inductive check, the EER over trials *not* in that subsample.

Protocol B (cross-corpus, balanced). Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA [18]; D2 (LibriSpeech bona fide vs. eight open TTS systems); In-the-Wild [17]; and the Arabic Audio Deepfake set [20]. Each held-out target’s source pool merges the other three with a 38-language MLAAD subset [19]; held-out target audio is never

TABLE I

OFFICIAL ASVSPOOF2021-DF EVAL, 400,435 TRIALS, 97.22% SPOOF, ON RELEASED XLS-R+AASIST CHECKPOINT SEED 2 [22]; BASELINE ARMS WERE RUN ON THIS CHECKPOINT ONLY. ALL ADAPTATION IS LABEL-FREE. DEFICIT $\delta = (100 - \text{EER}) - \text{acc}@0.5$ IS THE ACCURACY THE RANKING SUPPORTS BUT THE THRESHOLD DOES NOT DELIVER. ρ AND THE PREDICTION-RATE GAP ARE THE LABEL-FREE MONITORS OF SEC. II(D) — COMPUTABLE AT DEPLOYMENT, UNLIKE EVERY OTHER COLUMN.

	EER %	AUC	acc@0.5	deficit	ρ	gap
Source (as released)	4.51	.9923	98.87	-3.38	1.000	.000
+ median-threshold rule	4.51 [†]	.9923 [†]	59.06	—	—	—
+ ETA [8]	7.44 [‡]	.9288	98.93	-6.37	0.312	.007
+ Tent [7]	4.06 [‡]	.9765	98.52	-2.58	0.438	.013
+ SHOT [10]	6.38	.9725	77.64	+15.99	0.767	.225
+ TTA, fixed confidence 0.95	[8b]					
+ TTA, symmetric $q=0.3$	5.84	.9846	62.96	+31.20	0.816	.372
+ TTA, prior-corrected	4.02	.9931	98.95	-2.97	0.867	.007
(disjoint/inductive half)	4.03	.9931	98.95			
+ SAR [9]	4.47 [§]	.9923	98.88	-3.35	0.970	.000

[†] A threshold cannot change EER or AUC; both come from the score ROC. The median rule is a control for the *accuracy* claim and cannot produce any EER result in this paper. [‡] EER over a pool that is more than 93% a single tied score (Tent saturates 98.07% of scores to exactly 1.0, ETA 97.71%); read AUC. [§] SAR neither damages nor adapts: 308 recovery resets in 2,500 steps, AUC unchanged to four decimals.

in any source pool and six MLAAD languages are held out of every pool. Ten seeds per target per method.

Protocol C (a benchmark at its own prevalence). All 31,779 In-the-Wild clips at their native 37.2% fake rate — skewed the *opposite* way to DF’s 97.2%.

Comparison points. Source-only; the median-threshold rule; Tent, SHOT, ETA/EATA, SAR; AdaBN [14]; DANN [13] (trained to convergence with the held-out target’s unlabeled inputs — more privileged access than ours); an ASDG-style baseline [5]; and a target-supervised oracle. Every TTA arm adapts the same 16,706 parameters with the same lr, batch and pass count; none was tuned separately, and none could be, since per-target TTA tuning needs the target labels the setting assumes absent. Two published constants in the modern TTA baselines are degenerate at $C=2$ and were re-derived rather than transplanted. EATA’s redundancy filter ($|\cos(p, \bar{p})| < 0.05$) is calibrated for $C=1000$; on the binary simplex the minimum attainable cosine is 0.71 at $\bar{p}=(.5, .5)$ and 0.11 at $(.9, .1)$, so nothing passes until the running average itself reaches a vertex, after which only *minority*-predicted samples pass — neither is the stated mechanism, so we run the reliability filter alone, which is EATA’s own published ablation. SAR’s recovery trigger $e_0=0.2$ sits far below ImageNet’s $E_0=0.4 \ln 1000=2.76$ but *inside* the $C=2$ reliable band $H < 0.277$, firing on every batch and freezing the model at its initial weights (verified: 32 resets in 32 steps, zero net movement); we keep its relative position, $e_0 = (0.2 / \ln 1000) \ln C = 0.020$.

Implementation. 16 kHz audio, top-4-layer XLS-R fine-tuning for our own source models (Adam, $\text{lr} 2 \times 10^{-4}$, batch 32, 8 epochs). Adaptation: $\text{lr} 1 \times 10^{-4}$, $E=4$ passes, batch 32 (third-party crops matched to each checkpoint’s own training length). Code and manifests: [URL withheld].

IV. RESULTS

A. The class prior is the hidden variable

Table I is the main result, and Eqs. (1)–(2) anticipate it. Applied unmodified to a released checkpoint already better than the published Wav2Vec2-AASIST baseline here, the symmetric-quantile configuration *damages* it: accuracy $98.87 \rightarrow 62.96\%$, EER $4.51 \rightarrow 5.84\%$, manufacturing a 31-point deficit where there was none.

The published baselines are no better placed, and each fails through its own version of the assumption. SHOT minimises $\mathbb{E}[H(p)] - H(\mathbb{E}[p])$, whose diversity term is maximised when the mean prediction is *uniform* — at $C=2$ a hard prior of $\frac{1}{2}$, asking it to move 47 points of probability mass into the bona fide class; the regulariser introduced to *cure* Tent-style collapse encodes the assumption more literally than Tent does. ETA fails more instructively still: its reliability filter, meant to discard the high-entropy samples that destabilise entropy minimisation, admitted 99.8% of them, so ETA reduced to Tent and landed worse. An entropy-keyed filter separates confident from uncertain, not *confident-and-right* from *confident-and-wrong*, and mis-calibration manufactures exactly the latter. SAR is the exception that clarifies the rule: the only prior method that does not degrade the checkpoint, and it manages that by declining to move it (308 recovery resets in 2,500 steps, AUC unchanged to four decimals). So the claim is not that every published method fails here, but that *three of five prior configurations degrade a state-of-the-art detector at the standard benchmark’s own prevalence, the fourth avoids damage only by reverting its own updates, and none improves it.*

The prior-split budget reverses this: BBSE returns $\hat{\pi}=0.9715$ against a true 0.9708, comfortably inside Eq. (3), and EER improves to 4.02%. The change between the two arms is one scalar. Over three independently trained checkpoints (EER % / accuracy %):

	seed 2	seed 42	seed 240
source	4.51 / 98.87	4.56 / 98.78	3.82 / 99.11
+ symmetric q	5.84 / 62.96	4.08 / 61.80	4.66 / 65.24
+ prior-corrected	4.02 / 98.95	3.29 / 98.98	3.80 / 98.74

Two things replicate and one does not. The operating-point collapse under a symmetric budget holds on all three, as does prior-correction beating that budget ($-1.82, -0.79, -0.86$ EER). What does not is the symmetric budget’s effect on *ranking* (EER worse on two, better on one) — as Eq. (1) implies, a mis-specified budget corrupts the pseudo-labels and hence the operating point, while the ranking is inherited. Gains over source-only vary widely (1.27, 0.49, 0.02), so we rest on the unanimous controlled comparison.

B. The failure, and where the recipe already works

Table II is the same recipe at a prior that *satisfies* Eq. (2) ($0.372 > 0.3$). DeepFense s42 — the checkpoint that scores 4.56% on DF — arrives here ranking at AUC 0.917 while correct on 39.1% of clips, fewer than a majority-class predictor. Self-training repairs exactly that: on the three checkpoints

TABLE II

THE COMPLETE IN-THE-WILD BENCHMARK (31,779 CLIPS, PRIOR 0.372), FOUR RELEASED CHECKPOINTS ORDERED BY SOURCE CALIBRATION DEFICIT δ . EER % / ACCURACY %. THE SYMMETRIC BUDGET REPAIRS EVERY CHECKPOINT THAT HAS A DEFICIT; THE PRIOR-CORRECTED ONE DOES NOT, AND ITS SHORTFALL GROWS WITH δ — WHICH IS WHAT CORRUPTS THE PRIOR ESTIMATE (SEC. V).

checkpoint	δ_{src}	source	symmetric q	prior-corrected
DeepFense s2	2.1	9.64 / 88.3	9.71 / 85.7	9.89 / 87.1
ssl-aasist [21]	16.7	5.06 / 78.2	4.55 / 95.4	4.54 / 95.2
DeepFense s240	22.6	8.47 / 68.9	8.33 / 88.8	8.48 / 77.0
DeepFense s42	44.4	16.47 / 39.1	12.58 / 82.4	14.51 / 48.0

with a deficit it drives δ from 16.7, 22.6, 44.4 to 0.1, 2.9, 5.0; on the well-calibrated one it has nothing to repair and costs a little. On s42 it also raises AUC .917 $\rightarrow$.947 — which no threshold can do — with the disjoint half slightly better (12.27 / .949), so the ranking gain is not memorisation.

C. Why this is not a re-thresholding result

Both EER and AUC improve, and neither takes a threshold as input. The cleanest form of that argument uses the *same* prior estimate twice: re-point the threshold with it and take no gradient step, and accuracy moves 98.92 $\rightarrow$ 98.94% while EER stays at 4.51 by construction; feed it to the budget instead and EER goes to 4.02.

On *balanced* pools a one-line median-threshold rule is genuinely competitive with gradient adaptation on accuracy — we reported it as such, and it is the right control for a calibration claim. Resampling the DF scores across target priors shows why, and what it costs: the median rule gives 95.6% accuracy at a prior of 0.5 (shipped threshold 88.3, labelled oracle 95.7) and falls monotonically to 59.0% at the benchmark’s own 0.972, while a threshold at the $(1-\hat{\pi})$ quantile tracks the shipped one throughout.

That is the rule’s assumption, not an accident — and every cell on which it was previously evaluated satisfied that assumption by construction, having been balanced to a prevalence of 0.489–0.539 by the evaluation pipeline itself.

D. Cross-corpus, balanced pools: where the advantage disappears

Here the prior correction is inert by construction, so this arm asks what the rest of the method is worth when the assumption everyone makes happens to hold. All seven arms were run on identical folds (four leave-one-corpus-out targets, three seeds, twelve folds), mean EER over folds:

	ASV19	LS-TTS	ITW	Arabic	pooled
source	4.77	35.88	11.19	21.04	18.39
Tent [7]	22.16	34.07	23.89	48.89	32.25
ETA [8]	7.66	35.23	32.02	43.26	29.54
EATA [8]	14.76	38.55	25.31	46.75	31.34
SAR [9]	4.71	34.48	12.54	21.84	18.39
SHOT [10]	3.64	34.41	11.52	21.47	17.76
ours	3.76	34.64	11.27	20.11	17.44

Paired over the twelve folds we beat source-only (10/12, $p=0.027$), Tent ($p=0.012$), ETA ($p=0.003$), EATA ($p=0.016$) and SAR ($p=0.027$) — but **not SHOT** (17.44 vs. 17.76,

7/12, $p=0.47$). On balanced pools SHOT is our peer, and on ASVspoof2019 it is better. That is the point rather than an embarrassment: SHOT’s diversity term is a hard uniform prior, which costs nothing when the pool is balanced and destroys a released checkpoint when it is not (6.38% EER and 77.6% accuracy on DF, against our 4.02 and 98.95). The separation between the two methods lives entirely in the regime the field does not test. Tent, ETA and EATA are meanwhile unstable even here, at 29–32% pooled EER against a source-only 18.39, and SAR is again exactly inert (pooled EER identical to source-only to two decimals).

E. Label-free guards

Whether to keep an adapted model is itself decidable without labels. Over the eight adapted DF arms, rank agreement ρ and prediction-rate drift are complementary — ρ catches ranking collapse (ETA 0.31, Tent 0.44), the drift catches operating-point collapse (0.37, 0.39, 0.23) and is blind to the former — and requiring both separates the five harmful arms from the three harmless ones exactly, for any $\tau \in [0.70, 0.80]$. The plateau matters more than the exactness: a ρ -only rule at $\tau=0.85$, fitted on seven arms, looked equally exact and was falsified by the eighth, rejecting the study’s best result. Two thresholds on eight points is a diagnostic, not a rule; [stage 6/7] gives the leave-one-target-out version.

F. Breadth, and the mechanism

Over nine released checkpoints, five architectures and ten corpora (369 cells), the consistent effect is *re-calibration*: δ falls 7.29 $\rightarrow$ 1.96 (248/369, $p=2.2 \times 10^{-17}$), and where it lands barely depends on where it started (corr=+0.17). The mechanism is *not* learned invariance: a proxy $\mathcal{A}$ -distance and a linear domain-origin probe both *rise* after adaptation (Arabic 1.545 $\rightarrow$ 1.606), consistent with DANN underperforming while optimising that very quantity. Beyond any threshold the gradient step adds separability where the ranking is strong: within-attack-family AUC rises on all six ASVspoof2019 families (30/30, $p=10^{-9}$); on the AUC-0.71 target it is null.

V. LIMITATIONS

The prior is not identifiable, which bounds the correction. A label-free method sees only $\Pr[\hat{y}=1] = \pi \text{TPR}_{\mathcal{T}} + (1-\pi)\text{FPR}_{\mathcal{T}}$: one equation, three unknowns. BBSE closes it by assuming source error rates carry over — what covariate shift breaks — so an extreme prior and an extreme calibration failure look identical (DF/s2 predicts 97.4% fake at a true 0.972, ITW/s42 98.1% at a true 0.372). The estimate therefore degrades exactly as the operating point does — over seven cells and two corpora, $|\hat{\pi} - \pi|$ against the source deficit gives Pearson $r=0.998$ — and a badly wrong prior is worse than none (ITW/s42 12.58 $\rightarrow$ 14.51 EER). Estimating the prior and detecting that the estimate is wrong are one problem; two label-free detectors for it are falsified in our release.

Scope. The prior fixes the *budget*, not the threshold; the gain varies (1.27 to 0.02 EER across DF checkpoints); and a low-AUC start we cannot rescue at all.

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